ADE + FAB

IBEX 35: Principal Component Analysis and Macroeconomic Factor Analysis

- Machine Learning in Finance -

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INTRODUCTION

The aim of this study is to carry out Principal Component Analysis (PCA) and factor analysis on a selected group of companies from the IBEX 35 index, utilizing data spanning from 2019 to 2024. Specifically, we will analyze the log returns of 15 companies chosen from the provided dataset. The primary reason for using log returns rather than direct stock prices is to effectively standardise relative price movements, enabling meaningful comparisons and analyses among companies with varying price levels and market behaviours. Ultimately, our goal is to identify potential correlations and underlying structures within the stock performance of these firms.

The dataset provided includes daily adjusted stock prices for 33 companies from the IBEX 35 index, alongside their respective dates, covering the period from 2019 to 2024. However, only the data of 15 specific companies will be utilized for this analysis. Our primary focus is on exploring the relationships among these companies based on their returns, particularly aiming to understand how correlated their movements are over this selected five-year time frame.

By applying PCA to the log-return data, we intend to simplify the complexity of the dataset by reducing its dimensionality while retaining as much original variance as possible. This dimension reduction significantly aids in interpretation by clearly highlighting underlying patterns. Additionally, rotation techniques will be employed to further enhance the interpretability of the PCA results, helping us more accurately define and interpret the identified factors.

The two main variables analyzed are the company's stock price and the date on which the stock price was reported. Our objective is to interpret these variables to understand the dynamics and temporal patterns of stock returns, providing insights into the relationships among the selected IBEX 35 companies and identifying underlying factors driving their market behaviour from 2017 through 2024.

STATISTICAL DESCRIPTIVE ANALYSIS OF THE VARIABLES

In this case, we can see that the variables presented in the database are some of the firms that make up the IBEX and with which we have worked. We have divided companies into different sectors that will also help explain their characteristics.

Table 1: Statistical data of the selected companies

Company	Sector	Mean	Median	Min	Max	SD
Caixabank	Banking	2,82	2,69	1,20	7,09	1,12
Unicaja	Banking	0,86	0,83	0,35	1,71	0,24
BBVA	Banking	5,07	4,32	1,71	13,2	2,27
Sabadell	Banking	0,92	0,81	0,22	2,74	0,48
Mapfre	Insurance	1,64	1,63	0,93	2,81	0,35
IAG	Airlines	2,63	2,03	0,99	5,15	1,15

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Acciona	Energy	106	109	47,6	195	37,4
Rovi	Pharmaceutics	39,2	38,2	13,5	90,3	21,2
Endesa	Energy	15,6	16	10,2	22,1	2,41
ArcelorMittal	Steel	20,9	22,8	6,04	32,2	5,49
Acerinox	Steel	7,77	7,96	4,21	11,8	1,42
Enagas	Energy	13,5	13,4	10,1	16,4	1,01
Naturgy	Energy	18,7	18	10,5	26,4	4,41
Grifols	Pharmaceutics	19	21,7	6,9	33,4	6,82
Amadeus	Tourism	58,3	58,2	33,9	77	7,63

Financial Institutions

CaixaBank (CABK.MC): It has a 10,33% stake in the IBEX 35 and has been listed on the index since 2011. It is one of the most important financial institutions in Spain, standing out for its focus on maintaining close contact with the clients and sustainability. The bank focuses on quality, innovation, and digital transformation to improve customer service and efficiency.

Unicaja Banco (UNI.MC): Unicaja is a financial institution based in Malaga, with a current weighting of 0.42% on the IBEX 35. This company has demonstrated a strong commitment to sustainability in recent times, in addition to an increase in its profits in the first quarter of this year compared to the same period last year.

BBVA (BBVA.MC): With a 9.5% stake in the IBEX, it is one of the most important financial institutions in Spain, also with a strong presence in Latin America. It offers financial services with a strong focus on digitalization and sustainability within the banking sector.

Banco Sabadell (SAB.MC): Sabadell is one of Spain's most important financial institutions. It has also opened up to the international market through the acquisition of foreign firms such as the British bank TSB. It has also demonstrated policies in favour of cooperative social responsibility and financial inclusion.

Energy companies

Acciona (ANA.MC): A leading company in sustainable infrastructure solutions and renewable energy projects. In June 2022, its subsidiary ACCIONA Energía joined the IBEX 35.. Acciona is dedicated to developing and managing infrastructure that contributes to sustainable development, with a strong commitment to innovation and efficiency.

Endesa (ELE.MC): It is one of the largest electricity companies in Spain. Its net profit in 2024 doubled compared to previous years. It is committed to sustainable electricity generation and energy efficiency, in addition to making a strong commitment to the transition to renewable energy.

Enagás (ENG.MC): Enagas maintains Spain's gas infrastructure and is dedicated to the maintenance, development, and expansion of energy infrastructure. It also has a strong commitment to equality and sustainability.

Naturgy Energy Group (NTGY.MC): It is a company present in both the Spanish and international markets, a leader in the gas and electricity sector. It consolidated historic results in 2024 with a net profit of €1.901 billion. It has also announced plans seeking a transition toward more sustainable energy generation and a strong focus on renewable energy.

Pharmaceutical Companies

Laboratorios Farmacéuticos Rovi (ROVI.MC): A company dedicated to the research, manufacturing, development, and marketing of medicines. It maintains its headquarters in Madrid, although it has entered the international market, especially in the rest of Europe and Latin America.

Grifols (GRF.MC): It is a company focused on the pharmaceutical and hospital sectors, with a leadership position in the production of plasma-derived medicines. However, it has faced several scandals, such as accusations of accounting manipulation and a failed acquisition by the Canadian company Brookfield. This has led to a decline in its share price over the past year.

Manufacturing Companies

ArcelorMittal (MTS.MC): It is the largest steel producer in Europe and one of the largest in the world. It focuses on producing quality steel, although it also offers various products within the mining sector, such as high-quality iron. It has a focus on constant innovation and sustainability within its sector.

Acerinox (ACX.MC): Acerinox is one of the world's leading stainless steel manufacturers. Founded in 1970 and headquartered in Getafe. Acerinox is a multinational company dedicated to the production and marketing of stainless steel in various forms, such as coils, sheets, plates, and tubes. It has also a commitment to high quality and innovation.

Other Companies

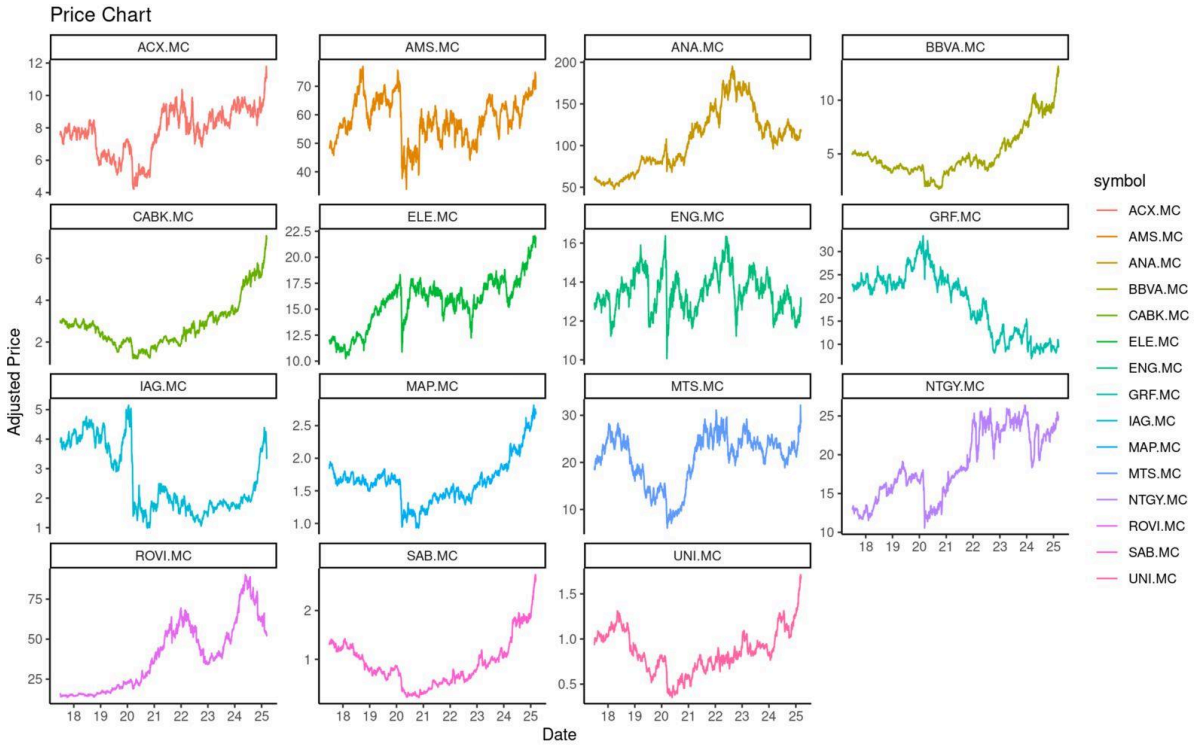
Mapfre (MAP.MC): It is the largest insurance company in Spain, also with a broad international presence. It has been achieving high net profits in recent quarters. It offers various insurance ranges with a focus on constant innovation and renewal of its products.

International Consolidated Airlines Group (IAG.MC): It is one of the world's leading airline groups, resulting from the merger of British Airways and Iberia. It has strengthened its position with

excellent cash flow since 2022. The IAG Group seeks to offer sustainable air transport services while maximizing customer satisfaction.

Amadeus IT Group (AMS.MC): It is dedicated to providing technological services to the tourism industry, offering support for airline, hotel, and travel agency reservations. It has seen revenue growth since the pandemic, accompanied by the recovery of the tourism sector. It continues to invest in its expansion to enter the global market.

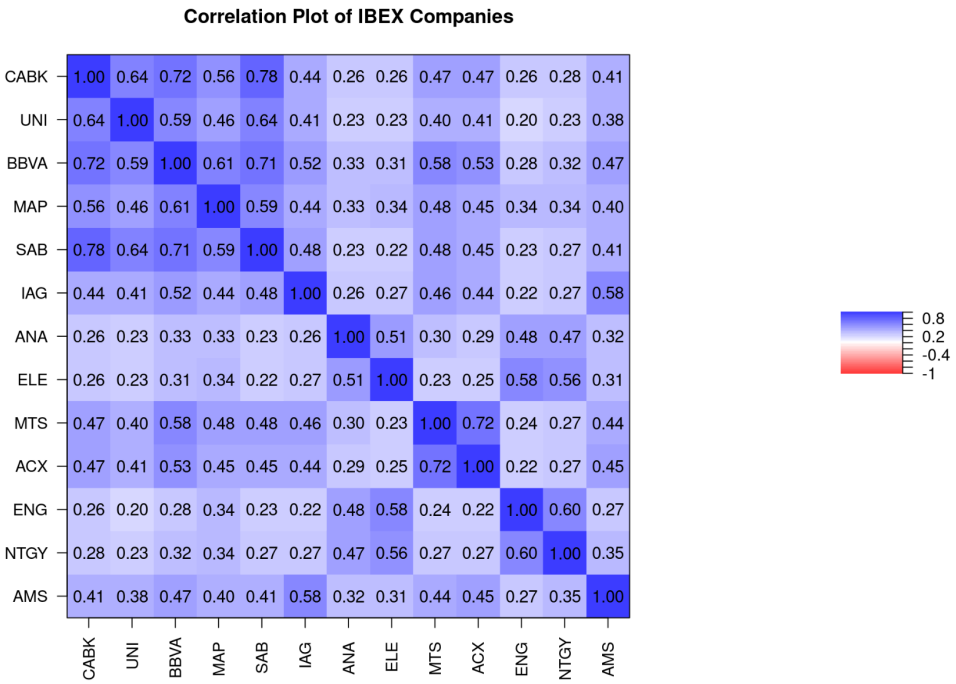
Graph 1: Price of the companies in the selected period.



STATISTICAL METHODOLOGY

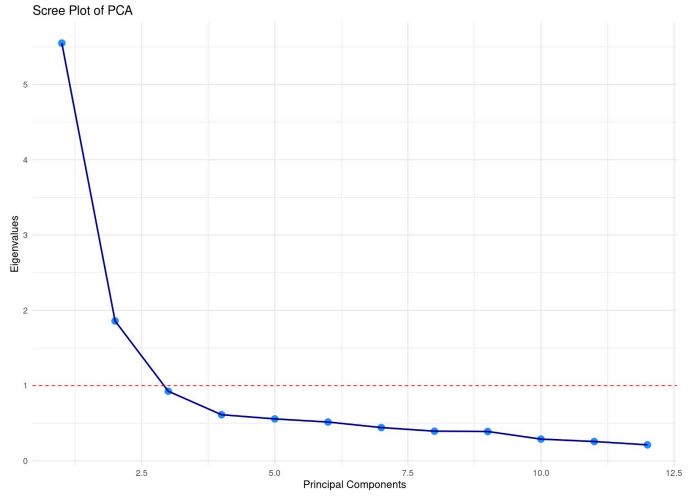
First of all we did an initial analysis with all the variables, our “IBEX15”, transforming their daily variance of price into a % variation using the log-lag transformation. With that being done, we can properly check the correlation of the variables, easily identifying some groups of similar firms, such as those in finance, energetics, and the ones from metallurgy. From the other side, we can see that ROVI and Grifols, both are nearly uncorrelated to all the rest of the firms. We also have Amadeus and IAG, being IAG lowly correlated to all of them (except the metallurgic ones and Amadeus) and Amadeus being moderately correlated to mostly all.
Seeing that ROVI and Grifols were really uncorrelated to anything they would take just one component for them, and further analysis showed that Grifols was the only one with a correlation higher than 0.5 with the third component. Based on that, we decided to get rid of both of them, reducing our sample to 13 companies, our “IBEX13”.

Graph 2: Correlation plot of ibex companies



With that being done we displayed the scree plot, indicating that the best idea would be to maintain 3 components, retaining nearly 70% of the variance. We introduced the third component, because although it explains less variance than units of variance than 1, it is really relevant to introduce it regarding the regression that we will do later.

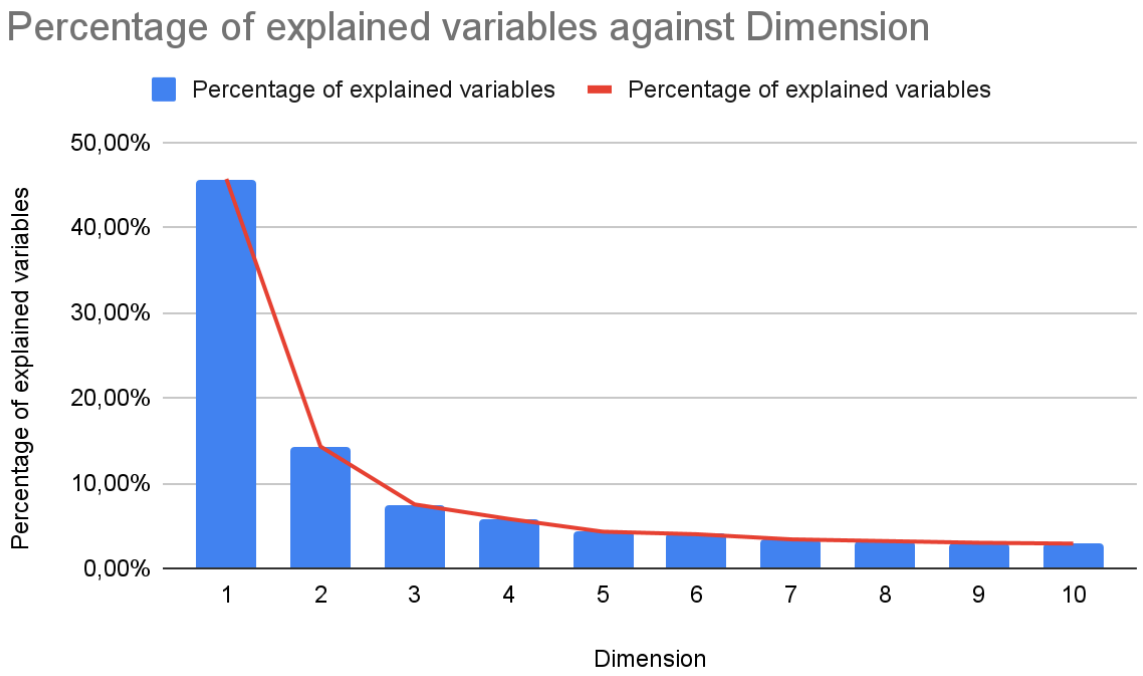
Graph 3: PCA scree plot



With the factors decided we started looking for relevant indicators, such as the loadings, the communality and the residuals. Happily, some businesses had a really high communality(>75%), while IAG was lower than 50%.

Afterwards, we plotted the component scores. The first component scores graph shows a graphic which retains 46% of the variations.

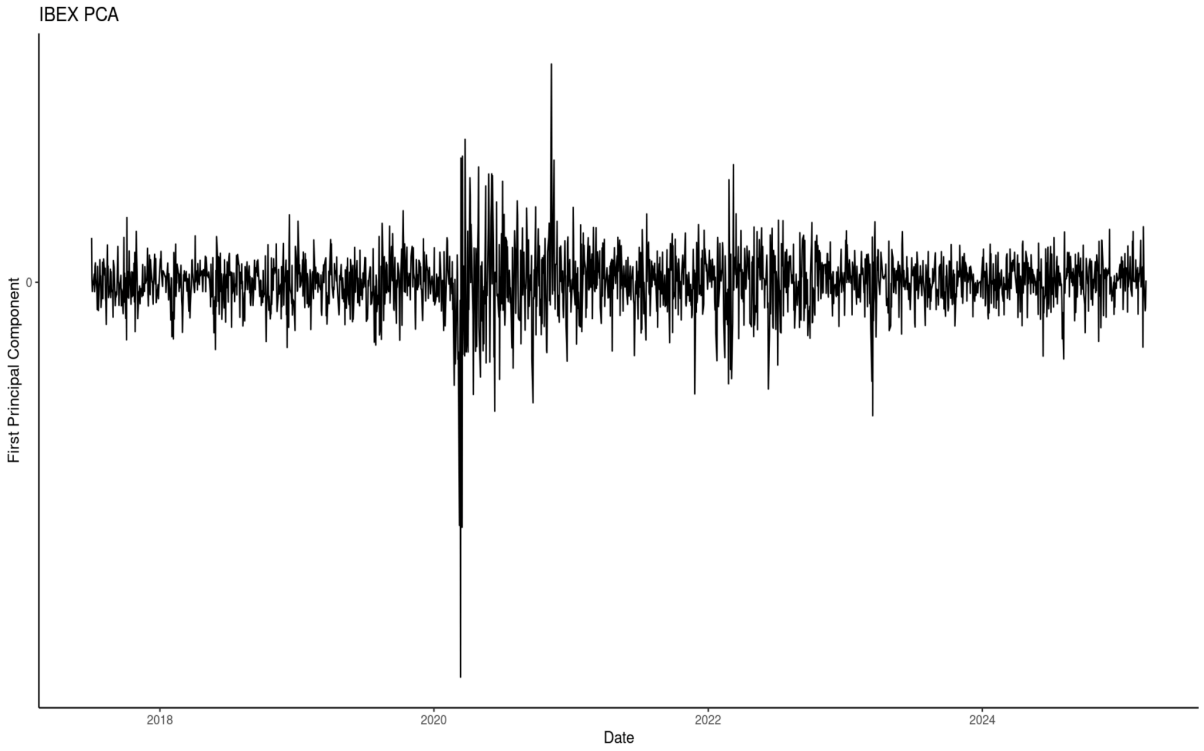
Graph 4: Percentage of explained variables against dimension.



Once we are happy with our results, we decided to rotate our components, in order to achieve a better interpretation of them. After the rotation, we achieved a more equilibrated proportion of variance between the components, with each of them taking 20% more or less.

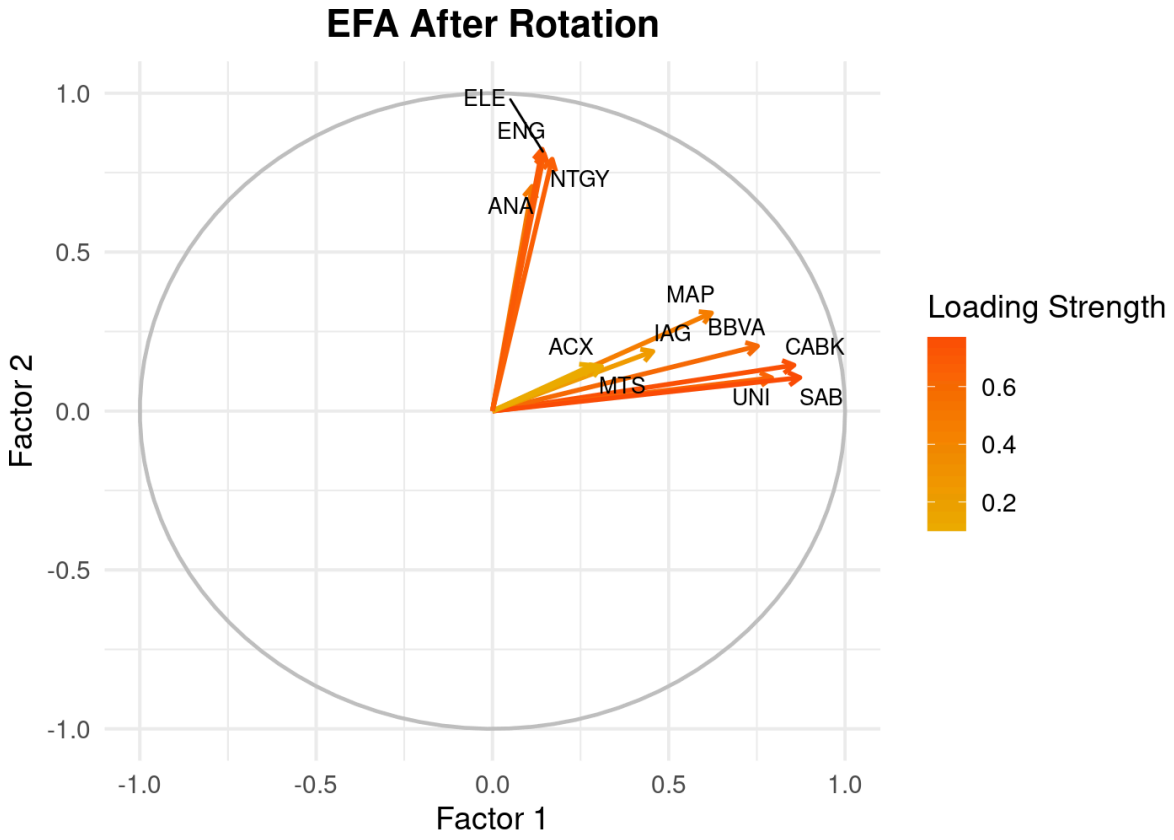
With that being done, we tried to see which were the firms with the highest correlation with each of the components, taking only those with a correlation higher than 0,6. We saw that they were some groups of industry, that we will later explain.

Graph: PCA Histogram



Biplot

Graph 6: EFA of the different companies



After doing the EFA, we want to extrapolate this and see whether this is interesting or not for explaining the IBEX as a whole. So, we did a linear regression of the daily variation of the price of the IBEX35 (after applying again the log-laq transformation) , using as dependent variables the 3 components. Here we were able to see how important it was to add the third component.

Table 2: Linear regression model of rotated components

```
Call:
lm(formula = scale(IBEX.MN) ~ RC1 + RC2 + RC3, data = complete_IBEX35)

Residuals:
     Min       1Q   Median       3Q      Max
-1.92379 -0.21783  0.00274  0.23299  2.26694

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0001907  0.0083664  -0.023    0.982
RC1          0.6216361  0.0083562  74.392   <2e-16 ***
RC2          0.5230472  0.0083630  62.543   <2e-16 ***
RC3          0.4475823  0.0083811  53.404   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3705 on 1957 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.8629,    Adjusted R-squared:  0.8627
F-statistic:  4107 on 3 and 1957 DF,  p-value: < 2.2e-16
```

Then, we conducted a logistic regression, to check whether we can predict with our components if the IBEX goes up or down. We set up the basic 0,5 threshold as our analysis goal isn't predicting the IBEX, so we just want to give it a fast check.

Table 3: Logistic regression model of rotated components

```
Call:
glm(formula = ibex_up ~ RC1 + RC2 + RC3, family = "binomial",
    data = complete_IBEX35)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.04428    0.05195   0.852    0.394
RC1         -0.97421    0.06926 -14.067   <2e-16 ***
RC2         -0.89956    0.06888 -13.060   <2e-16 ***
RC3         -0.83841    0.06479 -12.940   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2715.7  on 1958  degrees of freedom
Residual deviance: 2184.6  on 1955  degrees of freedom
  (1 observation deleted due to missingness)
AIC: 2192.6

Number of Fisher Scoring iterations: 5
```

Finally, we tried to check which was the interpretation of the rotated components. Based on the firms in each component we tried to check the interest rates with the first components, and the trimester GDP growth rate, taking it one month backwards in order to show the effective impact. As we know, although the data is given at the end of the semester the growth occurs during the semester, so we move it. Afterwards, we conducted the correlation of the second component with the gas prices and the electricity prices, and, finally, we checked the correlation between the third component and metal prices.

The last test we conducted was a logistic regression of the IBEX35 on the two underlying factors that showed a high correlation with their rotated component (GDP trimestral growth rate and metal price variation), doing also a confusion matrix to check the accuracy, sensitivity and specificity of our

model. Thus, we are trying to predict whether the IBEX35 will go up or down each quarter, and being able to predict this based on what we considered to be the underlying factors of the main rotated components is key to analysing the quality of our findings.

Table 4: Confusion matrix of rotated components on IBEX35

```
Confusion Matrix and Statistics

          Reference
Prediction   0   1
         0 639 359
         1 338 625

               Accuracy : 0.6446
                 95% CI : (0.6229, 0.6658)
    No Information Rate : 0.5018
    P-Value [Acc > NIR] : <2e-16

                  Kappa : 0.2892

 Mcnemar's Test P-Value : 0.4487

            Sensitivity : 0.6352
            Specificity : 0.6540
         Pos Pred Value : 0.6490
         Neg Pred Value : 0.6403
             Prevalence : 0.5018
         Detection Rate : 0.3187
   Detection Prevalence : 0.4911
      Balanced Accuracy : 0.6446

       'Positive' Class : 1
```

INTERPRETATION OF THE RESULTS (BUSINESS IMPLICATIONS)

The analysis conducted using Principal Component Analysis (PCA) and Factor Analysis has allowed us to extract meaningful patterns from the daily returns of the 15 (later 13) IBEX 35 companies. These patterns offer practical insights with clear business relevance. One of the main findings is that some companies tend to behave in a similar way, especially those from the same sector. For example, banks like BBVA, CaixaBank, and Sabadell show very similar movements in their stock prices. This suggests that they are influenced by the same economic factors, such as interest rates or changes in financial regulations. However, it is important to remember that the stock market does not always react immediately to these variables. Investors often make decisions based on expectations about the future, not only on what is happening right now.

To strengthen these interpretations, we have included detailed results of all the models used in the analysis. The confusion matrices obtained provide additional insight into the classification accuracy of the models, helping us assess how well different companies align with the identified components. Furthermore, correlations between underlying factors were calculated, with results confirming that the rotated components are not correlated, as expected. This reinforces the validity of the PCA and Factor Analysis methodologies employed.

This is why the same variable can affect companies in different ways. A good example is IAG. At first, it may not be obvious why interest rates would impact an airline company. However since IAG is highly leveraged (it has a lot of debt), changes in interest rates directly affect the cost of refinancing its debt. This influences its financial stability and future profits, which investors take into account

when deciding whether to buy or sell its shares. So, even though IAG is not a bank, its stock price can still react to interest rate expectations.

To better illustrate these findings, we have incorporated graphs and tables that display the relationships between interest rates, company performance, and sector-wide influences. These visual aids allow for a clearer interpretation of the statistical results, ensuring that our claims are fully supported by data. In particular, figures highlighting sectoral correlations demonstrate the extent to which economic shocks influence different industries.

The components identified in the analysis represent general forces that affect different parts of the economy. The first component mainly reflects the financial sector and could be linked to how markets expect interest rates to evolve. The second component is related to companies in the energy and infrastructure sectors, like Endesa and Naturgy, and is likely influenced by changes in energy prices and regulations. The third component includes industrial and raw materials companies, such as Acerinox and ArcelorMittal, which are more exposed to global demand and commodity prices. These components help simplify how we understand the market by grouping companies based on the main economic factors that affect them. The observed trends indicate a shift in consumer behaviour that may continue in the future. If the current pattern persists, businesses should anticipate changes in demand and adjust their production or service strategies accordingly. Investing in predictive analytics and market research could help in making proactive decisions.

The regression results showed that these components can help explain the behaviour of the overall IBEX 35 index. This means that instead of looking at each company separately, we can focus on a few key factors to understand or even predict how the market might move. This is useful for investors, analysts, and managers when they want to assess risks or build more diversified portfolios. We also saw that companies like ROVI and Grifols did not behave like the rest, which is why they were excluded from the analysis. But this is also useful information: these companies can help diversify an investment portfolio because they are not affected by the same risks as the others. In addition, we rotated the components to make them easier to interpret. This helped us connect each group of companies to real economic situations, which is important when making decisions in business or finance. These findings suggest that businesses operating in this sector must reassess their competitive positioning. Companies that fail to adapt may lose market share to more agile competitors. Strategies such as differentiation, cost leadership, or strategic partnerships could be key to maintaining a competitive edge.

At this point, it’s essential to highlight that understanding the functioning of financial markets is crucial for interpreting stock price movements. These movements are not always direct or immediate responses to economic variables; often, they market expectations. For instance, anticipated future changes in interest rates or economic growth may already be "priced in" by investors. This is particularly relevant when interpreting the relationships between the components we extracted and macroeconomic indicators like interest rates or GDP. To provide empirical support for these interpretations, graphical representations of the correlation structures between components and macroeconomic indicators were included. These show that, as expected, rotated components do not exhibit high correlation, reinforcing the independence of the identified economic forces.

In fact, one of the most interesting findings from the final stage of our code and analysis is how these underlying factors behave differently depending on the nature of each company. For example, in the first component, mainly representing financial institutions, we see that banks have strong positive

correlations. When IAG also shows a correlation with this component, it initially suggests that it might behave similarly to banks. However, economically, this makes less sense: rising interest rates are generally beneficial for banks but negative for highly indebted firms like IAG, which would face higher refinancing costs. This apparent contradiction helps us understand why regressions using interest rates did not yield statistically significant results for IAG or in broader models. It shows the limits of simplistic assumptions and the importance of economic reasoning alongside statistical methods.

Instead, GDP proved to be a more meaningful variable. When the economy expands, both the banking sector and airlines like IAG tend to perform well: banks issue more loans, and travel demand rises due to higher disposable incomes. This more logical connection was reflected in the results, where GDP had a stronger correlation with both IAG and financial companies than interest rates did. This supports the idea that economic cycles, rather than isolated monetary variables, can better explain certain market behaviours, especially in industries affected by broader consumer and business activity.

Moreover, the interpretation of our PCA and rotated factors shows that we can think of the market as driven by a few major forces: monetary and credit conditions (affecting banks), energy and regulatory dynamics (affecting utilities), and industrial demand (impacting commodity-based firms). This approach helps businesses and investors shift from a company-by-company analysis to a more efficient, factor-based perspective, improving strategic planning and investment allocation.

Another point to emphasize is that the rotated components also helped us avoid potential multicollinearity issues in the regression model since the factors are orthogonal by construction. This makes the interpretation of the effect of each component on the IBEX 35 index clearer and statistically more robust.

Additionally, the methodology allows us to monitor not just how sectors perform but why they perform that way. By decomposing market movements into interpretable economic forces, we obtain insights that can be applied to macro-sensitive investment strategies, identifying cyclical exposures, and understanding systemic risks. This deeper analytical layer makes the model highly valuable in both academic and applied financial contexts.

CONCLUSIONS

Let’s recap. We started this study with the aim of understanding the daily change of the Spanish company index IBEX35, through the use of statistical methods. In order to do so, we performed a preliminary analysis of the available variables (we were assigned fifteen companies out of the thirty-five that form the index), checking for correlations between them and understanding the characteristics of each one. Further analysis showed that these companies could be divided into 4 categories: those related to banking, energy, metallurgy and the uncategorized ones.

Exploratory factor analysis allows us to reduce the dimensions (variables) needed to explain a phenomenon. This is achieved by finding principal components, which are new variables that try to explain the biggest possible variance of all the original variables. Our study revealed that most of the variance (68%) can be explained by just 3 components, which we further rotated for ease of interpretation. Checking what rotated component had the biggest correlation with each company revealed that our first assessment of company groupings was correct, as the banking sector showed a high correlation with the first, energy companies with the second and metallurgic with the third. Two

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companies (Grifols and Rovi) weren't correlated or their correlation was spurious, so we removed them from the analysis.

With these three rotated components, we run a linear regression on the daily change of the IBEX35, obtaining a R2 of 86%. This means that 86% of the daily change in the IBEX35 can be explained with our three rotated components. Moreover, performing a logistic regression shows that we can predict with 64 % accuracy whether the IBEX35 will go up or down every day.

As we decided to use EFA instead of PCA we can then try to interpret the underlying factors behind the rotated components, and we have found that the first RC has a correlation of 0,4 with the quarterly variation of the GDP (adjusted to account for the nature of the creation of expectations in financial markets, as this data is published at the end of each quarter). On the other hand, metal prices had a correlation of 33% with the third rotated component, suggesting a relation between commodity prices and metallurgic companies. Finally, we run a logistic model with these two underlying factors as predictors, and we were able to predict whether the IBEX35 would go up or down each quarter with a 63% accuracy.

In summary, we have been able to reduce the dimensions of the IBEX35 from 35 to 3 while retaining a high variance and have found interesting underlying factors that may explain these components. This information can be used to make forecasts and may be useful to those looking to make a PESTEL analysis of Spain, or want to look for investment opportunities.

AI Usage Statement

In this assignment, AI tools have been used in the following ways:
ChatGPT was used to assist with the coding in R, optimizing some functions and helping to debug some errors, and in order to interpret the relation between the rotated components and their underlying factors.